Computational Insights into the Regulation of Gasdermin-Mediated Pyroptotic Cell Death

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Abstract

Pyroptosis is an inflammatory type of programmed cell death that plays a key role in innate immune responses and defending the host against infections. Gasdermin-D (GSDMD) plays as the central executioner of pyroptosis, being activated by proteolytic cleavage from inflammatory caspases. While atomistic molecular dynamics (MD) simulations have revealed the structural transitions of GSDMD during cleavage and pore formation, the regulation of GSDMD by post-translational modifications (PTMs), allosteric sites, and pharmacological agents remains poorly understood. This research proposal focus to establish a mechanistic framework for modulation of GSDMD-mediated pyroptosis using MD simulations, enhanced sampling, and machine learning (ML) to address: i) How PTMs affect membrane targeting, cleavage, and oligomerization ii) Allosteric sites that control conformational activation iii) Interactions with small molecules or antibodies for therapeutic inhibition. By integrating MD and ML approaches, this study will identify novel regulatory and druggable sites, providing predictive insight for rational modulation of pyroptosis in inflammatory and infectious diseases.

1. Scientific background

Pyroptosis is a form of programmed cell death characterized by its pro-inflammatory nature and morphological features reminiscent of necrosis [1, 2]. Initially described as caspase-1–dependent necrosis, pyroptosis is now recognized as an inflammasome-mediated form of programmed cell death [3, 4]. Inflammasomes are multiprotein signaling platforms that sense pathogen- and damage-associated molecular patterns and activate inflammatory caspases. These caspases—including caspase-1, -4, and -5 in humans, and caspase-11 in mice—initiate pyroptosis by cleaving downstream effectors, most prominently gasdermin D (GSDMD) [5].

GSDMD functions as the central executioner of pyroptosis. It comprises an N-terminal domain (NTD), which mediates membrane pore formation, and a C-terminal domain (CTD) that maintains autoinhibition under resting conditions [6]. Upon cleavage by inflammatory caspases, the liberated NTD oligomerizes and inserts into membranes to form pores, leading to membrane permeabilization, cell swelling, lysis, and the release of pro-inflammatory cytokines such as IL-1β and IL-18. While the NTD directly drives pore formation, the CTD serves as a negative regulatory element that prevents premature or aberrant pyroptotic signaling [7].

GSDMD activity is controlled not only by proteolytic cleavage but also by its intrinsic autoinhibitory architecture and post-translational modifications (PTMs) that influence protein stability, subcellular localization, and intermolecular interactions [8]. Emerging evidence highlights PTMs as critical upstream regulators of pyroptosis. Lipid modifications, such as palmitoylation, facilitate membrane targeting of cleaved GSDMD [9], whereas protein-based PTMs including acetylation, ubiquitination, and SUMOylation modulate GSDMD stability and abundance [10]. In particular, SUMOylation at lysine 177 is required for efficient pyroptotic signaling, whereas deSUMOylation by SENP3 destabilizes GSDMD and limits membrane damage [11].

Recent structural and atomistic MD studies have shown that small GSDMD-NT oligomers can form stable, lipid-stabilized, ion-conducting pores, clarifying the final membrane-disruptive phase [12]. However, pore formation represents only the terminal step; upstream PTM-dependent regulation ultimately determines GSDMD activity. While high-resolution structures provide valuable static snapshots, they cannot capture the dynamic transitions and allosteric pathways that underlie PTM-mediated control. Importantly, no current theoretical framework predicts how PTMs reshape GSDMD conformational ensembles or rewire its interaction network, leaving the mechanistic basis of pyroptotic tuning unresolved. Motivated by these gaps, this research proposal aims to employ theoretical approaches to elucidate how PTMs modulate GSDMD conformational pathways and guide the rational design of small-molecule modulators.

2. Research Objectives

The central research questions of this proposal are:

i. To explore how post-translational modifications (PTMs) affect membrane targeting, cleavage, and oligomerization of GSDMD.

ii. To identify allosteric sites that control the conformational activation of GSDMD.

iii. To investigate how interactions with small therapeutic molecules or antibodies modulate GSDMD activity.

To address these questions, this research program will employ an integrative computational framework combining advanced molecular dynamics simulations with machine learning based approaches. This strategy will enable a systematic dissection of GSDMD regulation at the atomic level, moving beyond static structures toward predictive models of inflammatory cell death control.

3. Materials and Computational methodology

3.1 System preparation and structural models

The high-resolution experimental PDB structures of full-length GSDMD, its NTD and CTD, and available GSDMD–caspase complexes will be used as starting models. Missing loops and unresolved regions in the PDB files will be reconstructed using homology modeling [13]

and structure prediction tools such as AlphaFold [14], followed by careful validation. Post-translational modifications of interest, such as SUMOylation and palmitoylation, will be explicitly modeled using validated classical force-field parameters. Both modified and unmodified GSDMD systems will be prepared to enable direct comparative studies.

3.2 Methodology

Conventional all-atom molecular dynamics (MD) simulations [15] will be performed to characterize the conformational dynamics, interdomain interactions, and stability of GSDMD under different regulatory states. To capture the rare and slow conformational transitions associated with post-translational modifications (PTMs) that are not accessible with conventional all-atom MD timescales, enhanced sampling techniques [16] such as accelerated MD [17] or metadynamics [18] will be employed. All simulations will be conducted using GROMACS with the CHARMM36 [19] force field to ensure accurate modeling of protein-lipid and protein-protein interactions.

Additionally, unsupervised machine learning (ML) methods such as clustering and dimensionality reduction will be used to simulation trajectories to identify dominant conformational states, transition pathways, and critical residues controlling activation. Therapeutic small molecules or antibody fragments will be docked to dynamically identified allosteric regions, followed by MD refinement to assess binding stability, induced conformational changes, and functional impact. Changes in conformational dynamics upon binding will be quantified to predict inhibitory or stabilizing effects on GSDMD activation, thereby revealing druggable or modulatory sites. The integration of MD, enhanced sampling, and ML will provide a mechanistic framework linking PTMs, allosteric regulation, and ligand interactions to GSDMD activation. The computational predictions can subsequently guide targeted mutagenesis or experimental validation, establishing testable hypotheses for rational modulation of pyroptosis in inflammatory and infectious diseases.

4. Expected outcome

This proposal addresses key gaps in understanding the dynamic regulation of GSDMD-mediated pyroptosis. By integrating molecular dynamics, enhanced sampling techniques, and machine learning analysis of PTM-induced conformational changes, this study will provide mechanistic insights into caspase recognition, allosteric regulation, NTD activation, and PTM-mediated modulation. The findings are expected to identify novel regulatory and druggable sites, guide the design of small therapeutic molecules or nanobodies, and enable rational modulation of pyroptosis in inflammatory and infectious diseases. Overall, this work will make a significant contribution to both mechanistic and translational understanding of inflammatory cell death.

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